LLM TRAINING PIPELINE GUIDE

What "Training a Model" Actually Means

Pretraining · Fine-tuning · RLHF · Distillation: four completely different processes, one overloaded word.

1. **Pretraining:** Builds foundation knowledge ($10M+, months)
2. **Fine-tuning (SFT):** Shapes instruction-following behavior ($50, hours)
3. **RLHF / DPO:** Aligns human preferences & safety (weeks)
4. **Distillation:** Compresses large teacher into small student (days)


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TERMINOLOGY DISAMBIGUATION

The Confusion This Solves

"We trained a model" can mean four very different things

When someone says they "trained" a model, they might mean:

- **Pretraining:** Millions of dollars over months on thousands of GPUs.
- **Fine-tuning:** Running a script on 500 examples for 20 minutes ($50).
- **RLHF / DPO:** Aligning behavior using human preference rankings.
- **Distillation:** Compressing a large model into a small one.

Disambiguation Branching Diagram

• Pretrain	$10M+ | Months | 1000s GPUs
• Fine-tune	$50 | Hours | 1 GPU
• RLHF/DPO	Varies | Weeks | Human Taste
• Distill	Varies | Days | 70B → 7B

Using the right word: and knowing which one you actually need: saves enormous time and money.


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PIPELINE ARCHITECTURE

The Standard Pipeline

How a modern chat model actually gets built

1. PRETRAINING

knows a lot, useless to talk to

Predicts text continuations (not a chatbot yet).

2. SUPERVISED FINE-TUNING (SFT)

Teaches the model to follow instructions and answer Q&A.

3. PREFERENCE TUNING (RLHF / DPO)

Makes the model helpful, safe, honest, and well-behaved.

4. (OPTIONAL) DISTILLATION

Transfers capability into a smaller, faster model.

Artifact Transformation Flowchart

```
[Raw Web Corpus (Trillions)] → Stage 1 → [Base Checkpoint]
[Instruction Pairs (10k)]    → Stage 2 → [Instruct Model]
[Preference Pairs A vs B]    → Stage 3 → [Aligned Chatbot]
[Teacher Synthesized Data]   → Stage 4 → [7B Edge Model]
```

Key Insight: A freshly pretrained model is NOT a chatbot. Everything after step 1 shapes *behavior*, not knowledge.

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STAGE 1 (FOUNDATION)

Stage 1: Pretraining

Learning language itself, from scratch

The model starts with random weights and learns by predicting the next token across an enormous corpus: trillions of tokens of web text, books, and code.

```
Input:  "The capital of France is"
Target: " Paris"
↓
adjust billions of weights slightly → repeat trillions of times
```

Self-Supervised Next-Token Training Loop

`[Raw Corpus] → [Tokenizer] → [Forward Pass] → [Cross-Entropy Loss] → [AdamW Weight Update]`

What makes pretraining distinctive:

- **Self-supervised:** No human labels needed; text provides answer.
- **Enormously expensive:** 1000s of GPUs, months, $millions.
- **Where knowledge comes from:** Facts, grammar, code absorbed here.
- **Frontier Labs only:** Essentially never done by individual teams.


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⚠ BASE MODEL LIMITATION

Base Model Behavior

Why pretraining alone isn't enough

A pretrained "base model" completes text: it doesn't answer questions. That distinction surprises people.

Prompt to BASE Model:

```
"What is the capital of France?"
```

Likely Output:

```
"What is the capital of Germany? What is the capital of Spain?
What is the largest city in Europe? ... "
```

↑ continuing a plausible list of questions, not answering!

⚖ Behavior Across Prompt Types

Prompt	Base Model	Instruct Model
Question	More questions	Direct Answer
Code	Completes file	Explains & implements

Why this happens: Training data contains countless quiz lists. Statistically, more questions is a reasonable continuation. The model has no concept of "being helpful" until later stages.


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STAGE 2 (INSTRUCTION)

Stage 2: Fine-Tuning (SFT)

Teaching the model to follow instructions

Continue training the pretrained model, but now on curated **instruction-response pairs** written or vetted by humans.

```
{"messages": [
  {"role": "user", "content": "Explain photosynthesis
simply."} ,
  {"role": "assistant", "content": "Plants use sunlight to
turn water ... "}
]}
```

SFT Target Token Loss Masking

```
User Tokens: [Explain ... ] → Loss Masked (0.0)
Assistant Tokens: [Plants ... ] → Loss Calculated (1.0)
```

What SFT changes:

- ✓ **Learns Assistant Format:** Question gets an answer, not continuation.
- ✓ **Learns Tone & Style:** Structure, tone, and formatting constraints.
- ✗ **No New Knowledge:** Does NOT add facts: those came from pretraining.

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DEVELOPER PRACTICAL GUIDE

Domain Fine-Tuning

The kind of fine-tuning realistic for your team

Taking an existing instruct model and further tuning it on your domain's data: the version of "training" most engineers will actually run.

When it helps:

- Consistent format/structure
- Domain tone & terminology
- Reduce long prompt headers
- High-volume narrow task

When it fails:

- **Adding facts** (Use RAG instead!)
- Anything unsolved by prompting

Architectural Decision Diagram

FINE-TUNING	RAG VECTOR SEARCH
Bakes behavior into weights	Injects dynamic context
"HOW to respond"	*"WHAT to know"*

RAG is 10x cheaper and provides real-time updateability without retraining risk!

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STAGE 3 (ALIGNMENT)

Stage 3: RLHF

Reinforcement Learning from Human Feedback

SFT teaches the model to imitate good answers. RLHF goes further: it teaches the model what humans **prefer** when several plausible answers exist.

The 3-Step RLHF Loop

1. Collect Preferences: Show humans 2 model outputs for same prompt → pick better one.

2. Train Reward Model: A separate model learns to predict human preference scores.

3. Optimize LLM (PPO): Adjust LLM so it generates outputs scoring high on reward model.

3-Model PPO System Architecture

```
[Actor LLM] → Output → [Reward Model Score] – (KL Penalty vs [Ref Model]) –> PPO Loss
```

Turns human taste and judgment into an optimizable reward function.

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RLHF EVALUATION

Why RLHF Matters

The stage that made chat models actually usable

What it achieves:

- Genuinely helpful answers
- Safety & harmful refusals
- Honest uncertainty tone
- Captures human taste

Difficulties:

- Reward hacking (verbose)
- Costly human labeling
- Unstable PPO training
- Encodes annotator bias

Reward Hacking Failure Mode

Models learn to write 500-word verbose essays with endless disclaimers because human annotators systematically rate longer answers higher, even when less accurate!

RLHF is why models refuse dangerous prompts and maintain polite conversation styles.


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MODERN ALIGNMENT

DPO: Modern Alternative

Direct Preference Optimization: same goal, far less machinery

DPO achieves preference alignment **without** training a separate reward model or running reinforcement learning at all.

Architectural Complexity Comparison

```
RLHF: Prefs → [Reward Model] → PPO RL → [Aligned Model] (4 models)
DPO:  Prefs → [Direct Implicit Loss] → [Aligned Model] (1 model!)
```

Why DPO is widely used:

- Simpler to implement & far more stable to train.
- Uses same preference pairs (chosen vs rejected response).
- Comparable alignment at lower complexity and cost.

Related variants: ORPO, KTO, and **RLAIF** (AI Feedback: an LLM provides preference labels instead of humans).


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STAGE 4 (COMPRESSION)

Stage 4: Distillation

Teaching a small model to imitate a large one

A large, capable **teacher** model generates outputs; a smaller **student** model is trained to reproduce them: inheriting capability at a fraction of the size.

Knowledge Transfer Pipeline

`TEACHER (70B) → Logits / Outputs → STUDENT (7B)`

Response Distillation: Student trains on teacher's final output text (simple API access).

Logit Distillation: Student matches full probability distributions (needs internal access).

Efficiency Impact: 7B student runs 10x faster and requires 10x less GPU RAM than 70B teacher!

DISTILLATION REALITY

Why Distillation Matters

Where most small models you use come from

- ✓ **Small models punch high:** Nearly every strong 3B-7B model is distilled from a larger model.
- ✓ **Reasoning transfer:** Large model reasoning traces teach small models to think step-by-step.
- ! **Inherits mistakes:** Student inherits teacher's capabilities AND biases/errors.
- **Legal Caution:** Many commercial API Terms of Service prohibit training competitor models from outputs.

Case Study: DeepSeek-R1 Distillation Pipeline

`[DeepSeek-R1 671B] → 800k Reasoning Traces → [SFT on Qwen 1.5B/7B/14B] → SOTA Small Models`

Distillation is the key engine powering efficient local edge models.


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PEFT INNOVATIONS

LoRA & QLoRA

How fine-tuning became affordable

Full fine-tuning updates every weight: requiring massive GPU RAM. **LoRA** freezes base weights and trains small "adapter" matrices instead.

Low-Rank Matrix Decomposition Math

`W_new = [Frozen Base W_0 (7B)] + [ΔW = Matrix A × Matrix B (~0.1% params)]`

```python
from peft import LoraConfig, get_peft_model
config = LoraConfig(r=8, lora_alpha=16, target_modules=
["q_proj", "v_proj"])
model = get_peft_model(base_model, config)
model.print_trainable_parameters() # "trainable: 0.12% of all
params"
```

VRAM Savings: Full 70B tuning requires 140GB+ VRAM (8x A100s). QLoRA 70B requires only 48GB VRAM (Single GPU)!


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COMPLETE MATRIX

Full Comparison

All four processes, side by side

Feature	Pretraining	SFT	RLHF/DPO	Distillation
Goal	Knowledge	Instructions	Preferences	Compress
Data	Trillions tokens	Q&A pairs	Rankings A/B	Teacher outputs
Cost	$Millions	$10s-$100s	Moderate	Moderate
Time	Months	Hours	Weeks	Days
Changes	Everything	Format/Tone	Safety/Style	Model Size
Who does it	Frontier Labs	Anyone	Labs/Teams	Labs/Teams

Bookmark this matrix: use it when planning your team's LLM customization strategy.


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SELECTION FRAMEWORK

Which Do You Need?

A decision guide for your own project

Project Decision Flowchart

```
Need FACTS? → Use RAG (Not Training)
Need FORMAT / STYLE? → Supervised Fine-Tuning (LoRA)
Need CHEAP EDGE MODEL? → Distillation
Need ALIGNMENT / SAFETY? → DPO
```

Cost-Saving Ladder: Always try Prompting → RAG → Fine-Tuning in that exact order!


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MYTH BUSTING

Common Misconceptions

What people get wrong about training

~~Fine-tuning adds facts~~
Unreliable for facts. Use RAG.

~~More data is better~~
High quality beats noisy pairs.

~~RLHF = Fine-Tuning~~
SFT imitates; RLHF optimizes.

~~Distill = Compression~~
Student is a separate model.

~~You must fine-tune~~
Prompting + RAG covers 90% apps.

~~Forgetting isn't real~~
Aggressive tuning degrades skills.

Avoid costly traps by starting with simple retrieval & clear prompts before fine-tuning!


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CORE TAKEAWAY

Four processes, one word

Pretraining builds knowledge. Fine-tuning shapes behavior. RLHF/DPO aligns preferences. Distillation compresses size.

PRETRAIN → Knowledge

FINE-TUNE → Behavior & Format

RLHF / DPO → Preferences & Safety

DISTILL → Size & Speed Efficiency

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